

串讲代码实现

C++ 基础

- C++ 版程序框架

```
1 #include<cstdio>
2 int main() {
3     int a, b;
4     while(scanf("%d%d", &a, &b) == 2)
5         printf("%d\n", a + b);
6     return 0;
7 }
```

```
1 #include<iostream>
2 #include<algorithm>
3 using namespace std;
4 typedef long long LL;
5 const int N = 110;
6 int a[N];
7 int main() {
8     ios::sync_with_stdio(false);
9     LL a, b;
10    while(cin >> a >> b) {
11        cout << min(a, b) << endl;
12    }
13    return 0;
14 }
```

- 引用

```
1 #include<iostream>
2 using namespace std;
3 void swapByValue(int a, int b) {
4     int t = a; a = b; b = t;
5 }
6 void swapByReference(int& a, int& b) {
7     int t = a; a = b; b = t;
8 }
9 int main() {
10    int a = 1, b = 2;
11    swapByValue(a, b);
12    cout << a << " " << b << endl;
13    a = 1, b = 2;
14    swapByReference(a, b);
15    cout << a << " " << b << endl;
16    return 0;
17 }
18 /*
19 1 2
20 2 1
21 */
```

```

1  int Abs(int n){ return n > 0 ? n : -n; }
2  double Abs(double n){ return n > 0.0 ? n : -n; }
3  printf("%d\n%.2f",Abs(-1),Abs(-2.1));
4  int Sum(int a = 0,int b = 0,int c = 0) {
5      return a + b + c;
6  }
7  printf("%d\n%d\n%d\n%d\n",Sum(),Sum(1),Sum(1,2),Sum(1,2,3));

```

- 结构体

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  struct Node{
6      int x, y;
7      bool operator < (const Node & b) const {
8          return x < b.x;
9      }
10 }a[N];
11 int main() {
12     int n; cin >> n;
13     for (int i = 1; i <= n; i++) {
14         cin >> a[i].x >> a[i].y;
15     }
16     sort(a + 1, a + 1 + n);
17     for (int i = 1; i <= n; i++) {
18         cout << a[i].x << " " << a[i].y << endl;
19     }
20     return 0;
21 }

```

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  struct Point{
6      int x, y;
7      Point(int u = 0, int v = 0) : x(u), y(v) {}
8      Point operator + (const Point& A) const {
9          return Point(x + A.x, y + A.y);
10     }
11 };
12
13 int main() {
14     Point a, b(1, 2);
15     Point c = a + b;
16     cout << c.x << " " << c.y << endl;
17     return 0;
18 }

```

- 模板

```

1  #include<iostream>
2  using namespace std;

```

```

3  template <typename T>
4  void mySwap(T &a,T &b) {
5      T t = a;
6      a = b;
7      b = t;
8  }
9  int main() {
10     int a = 1,b = 2;
11     mySwap(a, b);
12     cout << a << " " << b << endl;
13     double c = 3.1,d = 4.2;
14     mySwap(c, d);
15     cout << c << " " << d << endl;
16     return 0;
17 }

```

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  template <typename T>
6  struct Point{
7      T x, y;
8      Point(T u = 0, T v = 0) : x(u), y(v) {}
9      Point<T> operator + (const Point<T>& A) const {
10         return Point<T>(x + A.x, y + A.y);
11     }
12 };
13 int main() {
14     Point<int> a, b(1, 2);
15     Point<int> c = a + b;
16     cout << c.x << " " << c.y << endl;
17     Point<double> d, e(3.1, 4.2);
18     Point<double> f = d + e;
19     cout << f.x << " " << f.y << endl;
20     return 0;
21 }

```

STL初步

- 排序

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  struct Node{
6      int x, y;
7      bool operator < (const Node & b) const {
8         return x < b.x;
9     }
10 }a[N];
11 int main() {
12     int n; cin >> n;
13     for (int i = 1; i <= n; i++) {
14         cin >> a[i].x >> a[i].y;

```

```

15     }
16     sort(a + 1, a + 1 + n);
17     for (int i = 1; i <= n; i++) {
18         cout << a[i].x << " " << a[i].y << endl;
19     }
20     return 0;
21 }

```

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  struct Node{
6      int x, y;
7  }a[N];
8  bool cmp(const Node& A, const Node& B) {
9      return A.x < B.x;
10 }
11 int main() {
12     int n; cin >> n;
13     for (int i = 1; i <= n; i++) {
14         cin >> a[i].x >> a[i].y;
15     }
16     sort(a + 1, a + 1 + n, cmp);
17     for (int i = 1; i <= n; i++) {
18         cout << a[i].x << " " << a[i].y << endl;
19     }
20     return 0;
21 }

```

```

1  #include<iostream>
2  #include<algorithm>
3  using namespace std;
4  const int N = 110;
5  int a[N];
6  int main() {
7      int n; cin >> n;
8      for (int i = 1; i <= n; i++) {
9          cin >> a[i];
10     }
11     sort(a + 1, a + 1 + n, greater<int>());
12     for (int i = 1; i <= n; i++) {
13         cout << a[i] << endl;
14     }
15     return 0;
16 }

```

- 不定长数组: vector

```

1  #include<iostream>
2  #include<algorithm>
3  #include<vector>
4  using namespace std;
5  const int N = 110;
6  vector<int> a, b[N];
7  int main() {

```

```

8     a.push_back(1); a.push_back(2);
9     cout << a.front() << " " << a.back() << endl;
10    vector<int>::iterator it;
11    for (it = a.begin(); it != a.end(); it++) {
12        cout << *it << endl;
13    }
14    a.pop_back();
15    cout << a.size() << endl;
16    a.clear();
17    cout << a.empty() << endl;
18    return 0;
19 }

```

- 集合: set

```

1  #include<iostream>
2  #include<algorithm>
3  #include<set>
4  using namespace std;
5  const int N = 110;
6  set<int> s;
7  int main() {
8      s.insert(1), s.insert(2), s.insert(5);
9      if(s.find(1) != s.end()) puts("True");
10     set<int>::iterator it;
11     it = s.lower_bound(2); cout << *it << endl;
12     it = s.upper_bound(2); cout << *it << endl;
13     s.erase(it);
14     cout << s.count(5) << endl;
15 }

```

- 映射: map

```

1  #include<iostream>
2  #include<algorithm>
3  #include<map>
4  using namespace std;
5  const int N = 110;
6  string str;
7  int n, m;
8  map<string, int> h;
9  int main() {
10     cin >> n >> m;
11     for (int i = 1; i <= n; i++) {
12         cin >> str; h[str]++;
13     }
14     for (int i = 1; i <= m; i++) {
15         cin >> str;
16         if(h.find(str) != h.end()) cout << h[str] << endl;
17         else cout << 0 << endl;
18     }
19     return 0;
20 }

```

- 栈、队列、优先队列

```

1  #include<iostream>
2  #include<algorithm>
3  #include<queue>
4  using namespace std;
5  const int N = 110;
6  struct Node {
7      int x, y;
8      bool operator < (const Node & b) const {
9          return x < b.x || x == b.x && y < b.y;
10     }
11 }a[N];
12 priority_queue<Node> pq;
13 int main() {
14     int n; cin >> n;
15     for (int i = 1; i <= n; i++) {
16         cin >> a[i].x >> a[i].y;
17         pq.push(a[i]);
18     }
19     while(pq.size()) {
20         Node t = pq.top(); pq.pop();
21         cout << t.x << " " << t.y << endl;
22     }
23 }

```

C++11 简介

- auto

```

1  vector<int>::iterator it;
2  for (it = a.begin(); it != a.end(); it++) {
3      cout << *it << endl;
4  }

```

```

1  for (auto it = a.begin(); it != a.end(); it++) {
2      cout << *it << endl;
3  }

```

```

1  for (auto i : a) {
2      cout << i << endl;
3  }

```

- lambda表达式

```

1  vector<int> v;
2  sort(v.begin(),v.end(),[](const int &a,const int &b){return a > b;});

```

- unordered_set和unordered_map

参考资料

- C++ Primer Plus
- 算法竞赛入门经典
- 算法竞赛进阶指南
- oi-wiki.org